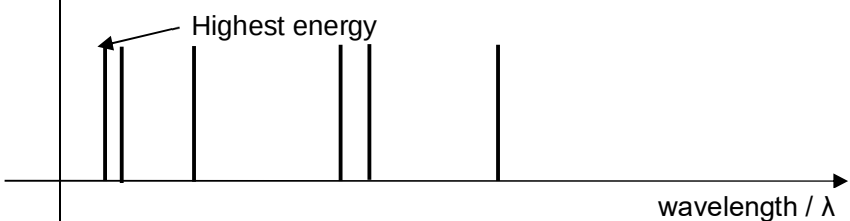


	7 10.7 11	[1] ans
6(a)	Work function energy refers to the minimum amount of energy needed to liberate an electron from its surface.	[1]
(b)	Energy of photon $E = \frac{hc}{\lambda} = (6.63 \times 10^{-34})(3 \times 10^8) / (540 \times 10^{-9})$ $= 3.683 \times 10^{-19} \text{ J} = 2.3 \text{ eV}$ Since the energy of the incident photon is lower than work function of metal, no electrons are emitted from surface.	[1] for substitution [1] for answer [1] for conclusion
(c)	If the wavelength of the radiation remains unchanged, <u>changes in intensity only changes the number of photons incident on metal surface per unit time per unit area.</u> Thus photon energy is still lower than work function, <u>no photoelectrons will be emitted.</u>	[1] [1]
(d)(i)	13.6 eV	[1]
(ii)	$-13.6 + 12.8 = -0.80 \text{ eV}$ Hence highest energy level that electrons can be excited to is -0.85 eV	[1] Must show working to get -0.80 eV

	Number of wavelengths = number of discrete transitions $= {}^4C_2$ $= 6$	[1] for correct answer
(iii)		[1] correct highest energy spectral line [1] three sections of spectral lines seen
(iv)	$\frac{hc}{\lambda} = (-0.85 - (-13.6)) \times 1.6 \times 10^{-19}$ $\lambda = 9.75 \times 10^{-8} \text{ m}$	[1] correct answer
7(a)	The root-mean-square (r.m.s.) value of an alternating current is	[2] or zero